Monopolistic Competition

Many sellers, and none of them selling quite the same thing

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Required readings

The core text is N. Gregory Mankiw, *Principles of Microeconomics*.

- **Chapter 17** — *Monopolistic Competition*.

Read it alongside these slides rather than instead of them.

Outline

1. **Where this sits** — four market structures from two questions.
2. **The three assumptions**, stated before anything is derived.
3. **Demand and marginal revenue** — why marginal revenue lies below price.
4. **The short run** — output, price, and the profit rectangle.
5. **What profits do** — entry, exit, and a demand curve that moves.
6. **The long run** — the tangency, derived.
7. **Against perfect competition** — excess capacity and the markup.
8. **Advertising and brand names.**
9. **Is it efficient**, and what the model leaves out.

PART 1

Where This Sits

Two questions sort every market

1. **How many sellers are there?** Many, a few, or one.
2. **Are the products identical?** A buyer who cannot tell one seller's output from another's is in a different market from a buyer who can.

Two questions, four answers. Every market structure economists name is one of the four, and each is a different combination of these two answers.

The four structures

	how many sellers	products	power over price
Perfect competition	many	identical	none
Monopolistic competition	many	differentiated	a little
Oligopoly	few	either	a lot
Monopoly	one	no substitute	the most

Read the top two rows against each other. The only thing that changes is whether the products are identical — and that alone is what gives the firm in the second row some power over its own price.

The two ends, restated

The price taker. Suppose the market price is \$34. This firm's own demand curve is horizontal at 34: it sells as much as it likes at 34 and nothing at all at 35, because an identical product is next door. One more unit adds exactly \$34, so $MR = P = \$34$, and $MR = MC$ reads $P = MC$.

The monopolist. There is nowhere else to buy. The firm's own demand curve is the market demand curve, and it slopes down. To sell more it must cut the price on *everything* it sells, so $MR < P$, and $MR = MC$ leaves $P > MC$.

Monopolistic competition takes the **downward-sloping demand curve** from the second and the **free entry** from the first.

Where most of the buying happens

Most of the economy you actually buy from lives in the second row.

- Restaurants, cafes and bakeries
- Clothing, shoes, furniture
- Hairdressers, dentists, plumbers, accountants
- Bookshops, gyms, hotels, software

Many sellers in each. No two of them offering quite the same thing. And nothing stopping a new one opening.

Active Learning 1

Put each of these in one of the four structures. **Name the structure and give the reason** — the reason is what is being marked.

1. Wheat farmers selling a standard grade at the market price.
2. The four firms that between them supply almost all the aircraft engines in the world.
3. Independent coffee roasters in a large city.
4. A water utility, the only supplier of piped water in its region.

Three minutes. Answer both questions for each one.

PART 2

The Three Assumptions

Three assumptions, stated before anything is built on them

1. **Many sellers.** Each one small relative to the market.
2. **Differentiated products.** The sellers' outputs are close substitutes but not identical.
3. **Free entry and exit.** No licence, no patent, nothing to stop a new firm opening or an old one closing.

Each of the three does a specific job later, and the next three pages say which. Nothing in this lecture arrives unannounced.

Assumption 1 — many sellers

Many sellers. Each firm is small enough relative to the market that its own decisions do not move any other firm's demand curve by an amount worth noticing.

If one roaster in a city of two hundred cuts its price, the other one hundred and ninety-nine each lose a scattering of customers. None of them notices, and none of them responds.

What it does later. There is no strategic interaction to model. No firm anticipates a named rival, so nothing here needs game theory. That is exactly what separates this market from oligopoly.

Assumption 2 — differentiated products

Differentiated products. The sellers' outputs are close substitutes but not identical. Some buyers genuinely prefer one seller's to another's.

The consequence, and it is the hinge of the whole lecture. Raise the price and this firm loses *some* customers, not *all* of them.

That is what a downward-sloping demand curve means. The firm faces one of its own — not a market price handed to it.

What it does later. Parts 3 and 4 are nothing but the consequences of this one assumption, and in them the firm behaves exactly like a monopolist.

Assumption 3 — free entry and exit

Free entry and exit. Any firm may open in this market, and any firm may close and leave it. No licence, no patent, no barrier of any kind.

Profit cannot persist where anybody may enter. If the firms here are making money, more firms open. If they are losing money, some close.

What it does later. Parts 5 and 6 are nothing but the consequences of this one assumption, and in them the firm behaves exactly like a competitive firm.

Active Learning 2

In each case, **name which of the three assumptions fails**, and say what the market becomes instead.

1. A city grants exactly twelve taxi medallions and issues no more.
2. Two hundred farmers sell an identical grade of milk into the same collection point.
3. Three airlines fly a route, and each publishes fares knowing the other two will match within the hour.
4. A drug is protected by a patent for twenty years.

Three minutes. Name the assumption, then name the market.

PART 3

Demand and Marginal Revenue

One firm, one demand curve

A coffee roaster in a city of many roasters. Its own blend, its own shop, its own customers. **Q** is thousands of pounds a year, **P** is dollars a pound.

$$P = A - 3Q$$

The 3 says that selling one thousand pounds more requires cutting the price by \$3. **A** is where the curve starts on the price axis, and it is the only thing in this lecture that ever moves. Take $A = 58$ for now.

Q	1	2	3	4	5	6	7	8
$P = 58 - 3Q$	55	52	49	46	43	40	37	34

Total revenue

Revenue is price times quantity, and the price comes from the row above it.

Q	1	2	3	4	5	6	7	8
P	55	52	49	46	43	40	37	34
$TR = P \times Q$	55	104	147	184	215	240	259	272

Every entry is the two numbers above it multiplied: $52 \times 2 = 104$, $40 \times 6 = 240$, $37 \times 7 = 259$.

Now compare the sixth column with the seventh.

What the seventh unit is worth

At six, the price is \$40 and revenue is \$240.

At seven, the price is \$37 and revenue is \$259.

The seventh unit adds \$19. It does not add \$37.

$$\underbrace{+\,37}_{\text{the new unit sells for 37}} \quad - \quad \underbrace{3 \times 6 = 18}_{\text{3 given up on each of the 6 already sold}} \quad = \boxed{19}$$

To sell the seventh the firm must charge 37 — **and it must charge 37 on all seven**, not only on the new one.

The whole column, and what it does

Q	0	1	2	3	4	5	6	7	8
P	58	55	52	49	46	43	40	37	34
TR	0	55	104	147	184	215	240	259	272
added by the unit	—	55	49	43	37	31	25	19	13

Read the price row and the bottom row down the columns together. **The price falls by 3 a step. The bottom row falls by 6 a step.**

Twice as fast — and it is twice as fast at every single step.

Marginal revenue, as a formula

Marginal revenue. The change in total revenue from selling one more unit.

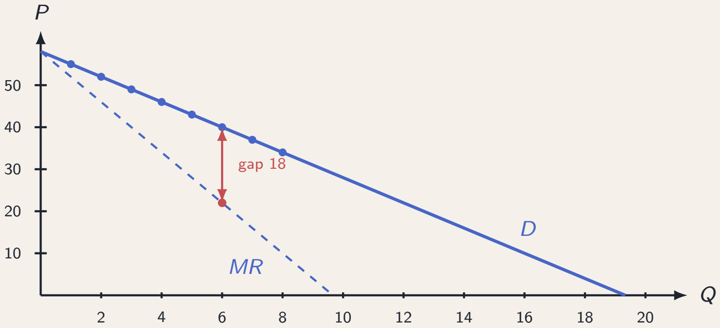
The column just built starts at 58 and falls by 6 a step, so:

$$P = A - 3Q \qquad \Longrightarrow \qquad MR = A - 6Q$$

Same intercept, twice the slope. At $A = 58$: $MR = 58 - 6Q$, so MR is 52 at one unit, 22 at six, and reaches zero at $Q = 58/6$.

MR lies below P everywhere except at the very first unit, where there is nothing already sold to give anything up on.

Demand and marginal revenue, drawn



Across: Q, thousands of pounds. Up: dollars a pound, with $A = 58$. Both lines start at 58. The marked gap at six units is $3 \times 6 = 18$: the price cut, on every unit already sold.

Reconciling the table with the line

From the table, the **seventh unit** adds 19. From the line, $MR = 58 - 6Q$ gives 22 at $Q = 6$ and 16 at $Q = 7$. These do not disagree.

The table measures revenue added **across** the step from 6 to 7, so its value belongs at the **middle** of that step: $58 - 6(6.5) = 58 - 39 = 19$. ✓

step	4th	5th	6th	7th
from the table	37	31	25	19
midpoint Q	3.5	4.5	5.5	6.5
$58 - 6Q$ there	37	31	25	19

Every step value lands on the line, at the midpoint of its step.

Active Learning 3

A different firm faces $P = 100 - 5Q$.

1. Find the price and total revenue at $Q = 4$, and again at $Q = 5$.
2. How much does the fifth unit add to revenue?
3. Get the same number a second way — the new price, less the price cut on the units already sold. **Both routes, and say why.**
4. Write down the marginal revenue equation for this firm.

Four minutes. Show every substitution.

PART 4

The Short Run

The cost side, once and for all

Fixed cost 144, variable cost $10Q + Q^2$. Average total cost is TC with every term divided by Q; marginal cost is the extra cost of one more unit.

$$TC = 144 + 10Q + Q^2 \quad ATC = \frac{144}{Q} + 10 + Q \quad MC = 10 + 2Q$$

Q	2	4	6	8	10	12	16
MC	14	18	22	26	30	34	42
ATC	84	50	40	36	34.4	34	35

This cost function never changes again.

The firm as we find it

The roaster is doing well. Its blend has a following, and its own demand curve is at

$$P = 74 - 3Q \qquad \text{so} \qquad MR = 74 - 6Q$$

Same slope as before. A higher intercept — at every quantity, this firm can charge \$16 more than the firm in Part 3 could.

Costs are unchanged and will stay unchanged. **Find the output, the price, and the profit.**

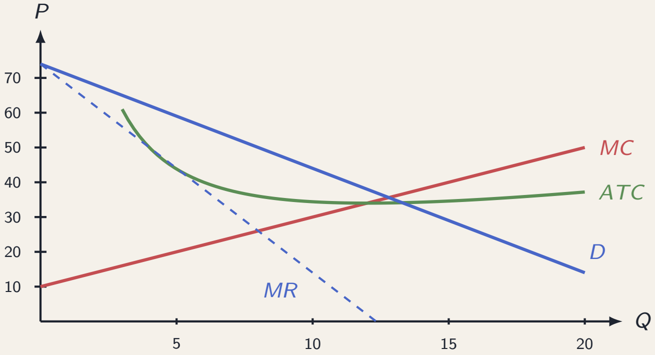
Step 1 — set marginal revenue equal to marginal cost

Produce one more unit while it adds more to revenue than it adds to cost. Stop where the two are equal.

$$\underbrace{74 - 6Q}_{MR} = \underbrace{10 + 2Q}_{MC}$$

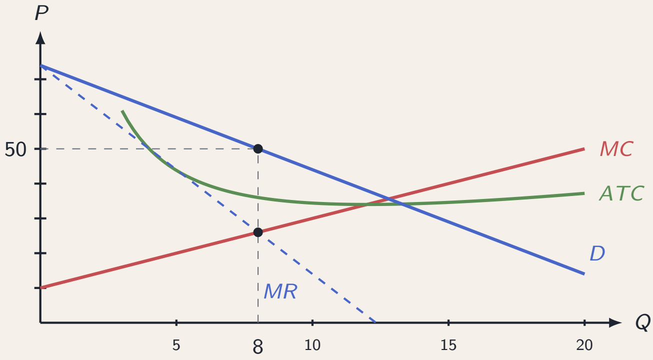
$$74 - 10 = 2Q + 6Q \quad\Longrightarrow\quad 64 = 8Q \quad\Longrightarrow\quad Q = 8$$

Check: at $Q = 8$, $MR = 74 - 48 = 26$ and $MC = 10 + 16 = 26$. ✓



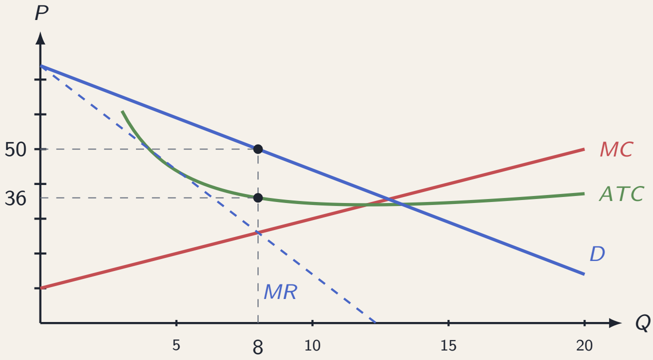
Across: Q, thousands of pounds. Up: dollars a pound. $A = 74$, and nothing is marked yet.

Step 2 — read the price up to the demand curve



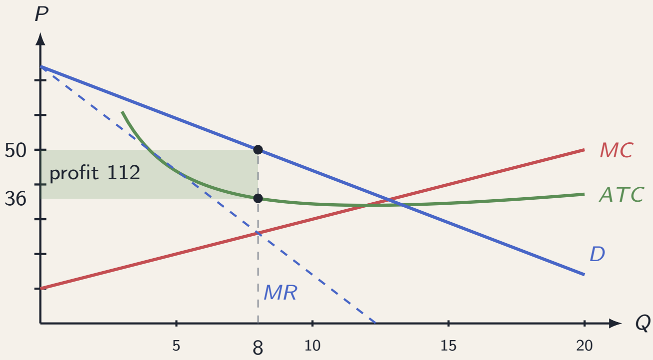
$MR = MC$ fixes the *quantity* at 8. The *price* is what buyers will pay for 8, which is on the demand curve: $P = 74 - 3(8) = 74 - 24 = 50$.

Step 3 — read the cost per unit up to average total cost



$$ATC(8) = \frac{144}{8} + 10 + 8 = 18 + 10 + 8 = 36.$$ The firm sells at 50 and each unit costs it 36 on average.

Step 4 — the profit rectangle



Height $50 - 36 = 14$ per unit. Width $Q = 8$. Profit $= 14 \times 8 = 112$.

The error worth naming

Never read the price off the marginal revenue curve.

At $Q = 8$ the marginal revenue curve is at $74 - 6(8) = 74 - 48 = \mathbf{26}$.

That is *not* a price. **26 is what the eighth unit adds to revenue** — the 50 it sells for, less the \$3 given up on each of the 7 already sold, which is $50 - 21 = 26$. ✓

Nobody pays 26. Every buyer pays **50**. Reading 26 as the price understates revenue by $24 \times 8 = 192$ and turns a profit of 112 into an apparent loss.

Active Learning 4

Same costs: $MC = 10 + 2Q$ and $ATC = \dfrac{144}{Q} + 10 + Q$. A different roaster faces $P = 82 - 3Q$.

1. Write down MR, then solve $MR = MC$ for Q.
2. Find the price. **Say which curve you read it off and why.**
3. Find ATC at that quantity, and the profit.
4. Check the profit a second way, from total revenue less total cost.

Five minutes. The reason in part two is worth as much as the arithmetic.

PART 5

What Profits Do

A profit is a signal, and somebody reads it

The roaster is making \$112,000 a year above what its owners could earn elsewhere. **Nothing prevents anyone else opening a roastery.**

So they do. That is the whole content of the free-entry assumption, and it is the only thing driving the rest of this lecture.

Each new roaster opens with its own blend and takes some customers from the ones already there — **including this one**.

Entry moves this firm's demand curve

Fewer customers at every price. That is a demand curve shifting left.

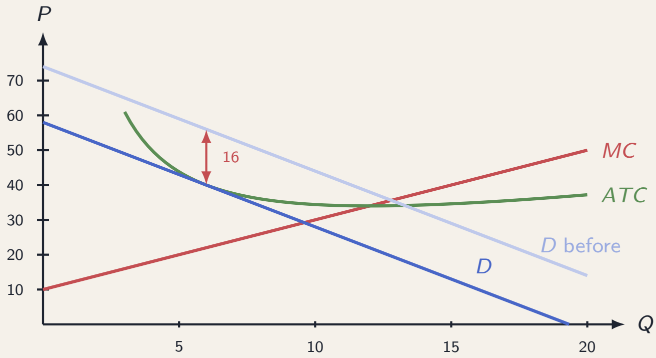
> **Same thing, read the other way.** To sell any given quantity, this firm must now charge less — because some of the people who would have bought that quantity are buying somewhere else.

$$P = 74 - 3Q \qquad \longrightarrow \qquad P = 58 - 3Q$$

A fall of **16 dollars at every quantity**. At 6 units the firm could charge $74 - 18 = 56$ before, and $58 - 18 = 40$ after.

Costs have not moved. Part 6 works out where the shifting stops.

The shift, drawn



The cost curves have not moved. Only the demand curve has, and the marked distance is the 16 dollars it fell by — the same at every quantity.

Losses run the argument backwards

Suppose instead the curve had gone too far, to $P = 42 - 3Q$.

$MR = MC$	$42 - 6Q = 10 + 2Q \Rightarrow 32 = 8Q \Rightarrow Q = 4$
price, off demand	$P = 42 - 3(4) = 42 - 12 = 30$
average total cost	$ATC(4) = \frac{144}{4} + 10 + 4 = 36 + 14 = 50$
profit	$(30 - 50) \times 4 = -80$

A loss of \$80,000 a year. So some roasters close. Their customers go to the survivors, and each survivor's demand curve shifts **right**.

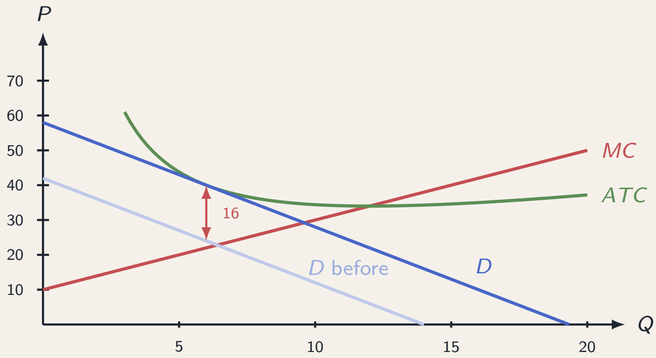
Does it shut down?

A loss is not automatically a reason to stop producing. **The fixed 144 is owed either way in the short run.** Compare the two options.

	produce 4	**produce nothing**
revenue	$30 \times 4 = 120$	0
variable cost	$10(4) + 4^2 = 56$	0
fixed cost	144	144
result	-80	-144

Producing loses 80; closing the doors loses 144. So it keeps producing — the price of 30 covers the $14 average variable cost.

Exit, drawn



The pale line is the loss-making curve at 42; the solid one is after some rivals have closed.

Same mechanism, opposite direction.

Active Learning 5

The roaster in Active Learning 4 faced $P = 82 - 3Q$ and made a profit of 180.

1. **Before calculating anything**, predict which way its demand curve moves, and say why.
2. Now suppose the movement stops at $P = 58 - 3Q$. By how much has the intercept fallen?
3. At 9 units, what could the firm charge before, and what after?
4. One sentence: what stops the movement going further?

Four minutes. Do part one before you touch a pen to the arithmetic.

PART 6

The Long Run

What has to be true when the movement stops

Entry continues while profit is positive. Exit continues while profit is negative. So the movement stops where profit is exactly zero.

Profit is $(P - ATC) \times Q$, and Q is not zero, so:

$$\text{profit} = 0 \qquad \iff \qquad \boxed{P = ATC}$$

But the firm is still choosing its output, so $MR = MC$ must hold as well. **Two conditions, and both have to be true at the same quantity.**

Condition 1 — marginal revenue equals marginal cost

Leave A as an unknown and solve as before.

$$\underbrace{A - 6Q}_{MR} = \underbrace{10 + 2Q}_{MC}$$

$$A - 10 = 8Q \quad \Longrightarrow \quad \boxed{A = 8Q + 10}$$

One equation linking the two unknowns. Check it against Part 4: with $Q = 8$ it gives $A = 74$. ✓

Condition 2 — price equals average total cost

Price comes off the demand curve, and average cost from the cost function.

$$\underbrace{A - 3Q}_{P} = \underbrace{\frac{144}{Q} + 10 + Q}_{ATC}$$

Add $3Q$ to both sides to get A on its own:

$$\boxed{A = \frac{144}{Q} + 10 + 4Q}$$

A second equation in the same two unknowns. Check it against Part 5: with $Q = 4$ it gives $36 + 10 + 16 = 62$, not 42 — and the firm at $A = 42$ was **losing** money, so it should fail this one. ✓

Solve them together

Both expressions equal A, so they equal each other:

$$8Q + 10 = \frac{144}{Q} + 10 + 4Q$$

The 10 appears on both sides and cancels. Take $4Q$ from both sides, then multiply by Q and divide by 4:

$$4Q = \frac{144}{Q} \implies 4Q^2 = 144 \implies Q^2 = 36 \implies Q = 6$$

And then $A = 8(6) + 10 = 58$. **The long-run demand curve is** $P = 58 - 3Q$, and it was not chosen. It was solved for.

Both conditions, checked at $Q = 6$

Condition 1	$MR = 58 - 6(6) = 58 - 36 = 22$	$MC = 10 + 2(6) = 10 + 12 = 22$
Condition 2	$P = 58 - 3(6) = 58 - 18 = 40$	$ATC = \frac{144}{6} + 10 + 6 = 24 + 16 = 40$

Both hold, at the same quantity. Marginal revenue meets marginal cost at 6, and at that very quantity price meets average total cost.

Profit $= (40 - 40) \times 6 = \mathbf{0}$. Nobody enters, nobody leaves.

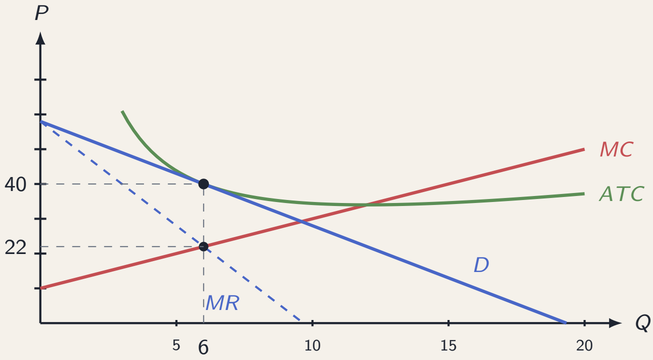
The same result, from the profit table

With $P = 58 - 3Q$ and $TC = 144 + 10Q + Q^2$, compute profit at each quantity:

Q	3	4	5	**6**	7	8	9
P	49	46	43	**40**	37	34	31
TR	147	184	215	**240**	259	272	279
TC	183	200	219	**240**	263	288	315
profit	-36	-16	-4	**0**	-4	-16	-36

The best the firm can do is exactly nothing, and it does it at 6. Any higher demand curve and the peak would be positive, so entry would continue. Any lower and it would be negative, so firms would leave.

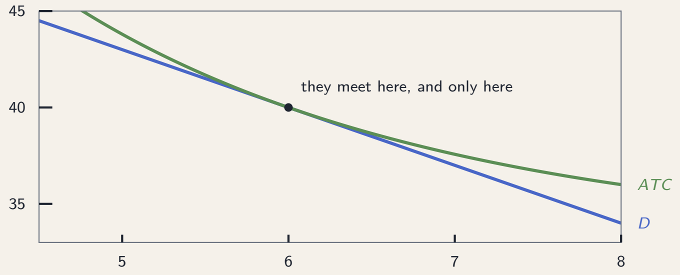
The long run, drawn



Upper dot: demand *touches* average total cost, price 40 and cost 40, so no rectangle.
Lower dot: $MR = MC$ at 22. Both above the same quantity, 6.

Why they touch rather than cross

The same two curves, magnified. Q from 4.5 to 8, P from 33 to 45.



ATC is above demand everywhere except at 6. At 5 it is 43.8 against a price of 43; at 7, 37.6 against 37. **The firm loses money at every quantity but one.**

Active Learning 6

A different market. Same marginal cost, $MC = 10 + 2Q$, but a fixed cost of **100** instead of 144, so $ATC = \dfrac{100}{Q} + 10 + Q$. Demand still has slope -3: $P = A - 3Q$.

1. From $MR = MC$, write A in terms of Q.
2. From $P = ATC$, write A in terms of Q again.
3. Set them equal and find the long-run Q, then A, then P.
4. Check **both** conditions hold at your quantity.

Five minutes. Part four is the part that matters.

PART 7

Against Perfect Competition

The same firm, if the product were identical

Keep the cost function. Take away the differentiation, so the firm is a price taker facing a horizontal demand curve.

- Free entry still drives profit to zero, so $P = ATC$.
- A price taker has $MR = P$, so $MR = MC$ reads $P = MC$.
- Both at once means $MC = ATC$ — and that happens only at the **bottom** of ATC, at $Q = 12$ where both are 34.

price	$P = 34$, and $34 = 10 + 2Q$ gives $Q = 12$
profit	$TR = 34(12) = 408$; $TC = 144 + 120 + 144 = 408$; profit $= 0$

Same costs, same zero profit, **different quantity and different price.**

The efficient scale, without calculus

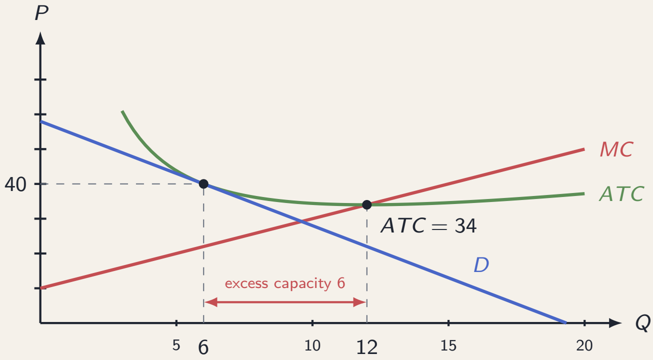
$ATC = \dfrac{144}{Q} + 10 + Q$. The 10 is fixed, so the question is where $\dfrac{144}{Q} + Q$ is smallest.

Q	8	9	**12**	16	18
$144/Q$	18	16	**12**	9	8
$+\,Q$	8	9	**12**	16	18
sum	26	25	**24**	25	26
ATC	36	35	**34**	35	36

The two terms trade off: one falls as output rises, the other grows. The sum is least where they are equal, which is $144/Q = Q$, so $Q^2 = 144$ and $Q = 12$.

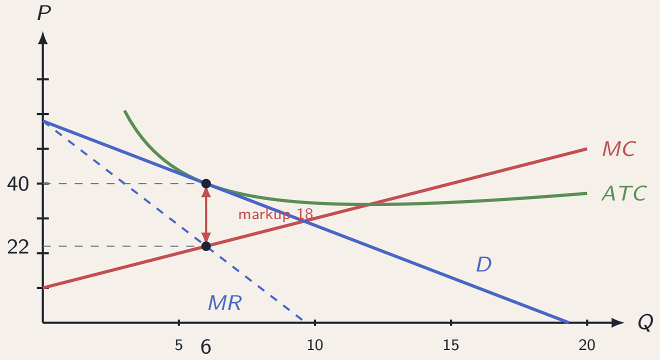
Efficient scale: the output at which ATC is lowest.

Difference 1 — excess capacity



The firm produces **6**; average cost is least at **12**. It operates on the falling part of its cost curve. A competitive firm with these costs produces 12, at the bottom.

Difference 2 — the markup



Price 40, marginal cost 22. The firm charges $18 above what the last unit costs to make.

A competitive firm charges 34 and its last unit costs 34 — **a markup of nothing**.

The two structures side by side

same costs, both in long-run equilibrium	perfect competition	monopolistic competition
the firm's own demand curve	horizontal	downward-sloping
quantity	12	6
price	34	40
average total cost	34	40
profit	0	**0**
marginal cost	34	22
markup, $P - MC$	0	18
excess capacity	0	6

Both make zero profit. Every other difference traces to the first line. Tilt the demand curve and MR separates from P; the firm stops short of the bottom of ATC, and price ends up above marginal cost.

Active Learning 7

Return to the firm in Active Learning 6: fixed cost **100**, $MC = 10 + 2Q$, long-run $Q = 5$ at a price of **35**.

1. Find its efficient scale. **Use the two-terms-equal argument**, not a formula you have memorized.
2. What is its excess capacity?
3. What is its markup over marginal cost at $Q = 5$?
4. A price-taking firm with the same costs: what price and quantity, and what profit?

Five minutes. Show the substitution in every part.

PART 8

Advertising and Brand Names

Why advertising belongs here

A firm selling an identical product has nothing to advertise.

A wheat farmer who buys a billboard does two things: raises demand for wheat in general, which helps every farmer equally, and pays for it alone. Nobody does that.

A firm with its *own* product captures the gain from telling people about it. **Advertising is a feature of differentiated markets**, and it is where most advertising in the economy happens.

The case against

1. **It manipulates tastes rather than informing.** Much of it carries no information about price, ingredients or performance. It associates a product with a feeling.
2. **It raises barriers to entry.** If customers must be won by advertising, a newcomer must spend before it can sell, and that is an obstacle an incumbent has already cleared.
3. **It is a cost that produces nothing.** The resources go into moving customers between sellers, not into making anything.

Each of the three is an argument that advertising **reduces** welfare.

The case for

1. **It conveys information.** That the product exists, what it costs, where to buy it, and what it does. A buyer who does not know a shop exists cannot choose it.
2. **It makes entry *possible*.** A firm nobody has heard of has no other way of being heard of. Without advertising, the advantage would sit entirely with whoever is already known.

Notice that the second point contradicts the second charge directly. Advertising is a cost of entry and it is also the means of entry, and which effect dominates is not settled by argument alone.

The signalling argument

The content of an advertisement may say nothing useful. The willingness to spend on it can still be informative.

Why. Advertising is worth paying for only if the customers it wins *come back*. A firm that expects repeat purchases can recoup the cost. A firm selling something poor makes one sale and cannot.

So a buyer who sees an expensive campaign can reason: **whoever paid for this expects me to come back**. And a firm that did not expect that would not have paid.

The signalling argument, in numbers

A customer won is worth $6 in profit *per purchase*. A good product is bought **10** times by each customer; a poor one is bought **once**.

	good product	poor product
purchases per customer	10	1
a customer is worth	$60	$6

A campaign that wins one customer and costs anything **between $6 and $60** is worth running for the good firm and not for the poor one. At $20, only the good firm advertises — so the buyer has learned which kind of firm it is, from an advertisement that said nothing.

Brand names, on the same argument

A brand name is a **stake the firm has put down**, and it works for the same reason a campaign does.

The reason. A name worth protecting gives the firm a reason to maintain quality: disappointing one customer costs it every future sale under that name. A firm with no name to protect bears none of that.

So the incentive to keep quality up rises with how much the name is worth.
That is why a chain enforces standards across shops it will never visit — and why a stall with no name need not.

Active Learning 8

A customer won is worth \$8 in profit per purchase. A good product is bought **12** times by each customer; a poor one is bought **twice**.

1. What is a customer worth to each kind of firm?
2. For which campaign costs does advertising separate the two?
3. At a cost of \$50, what does a buyer learn from seeing the campaign?
4. Now suppose the product is bought **once in a lifetime** by everyone. Does the argument still work? **Why.**

Four minutes. Part four is the one worth thinking about.

PART 9

Is It Efficient, and What the Model Leaves Out

The trade that does not happen

In the long run the firm sells 6 at a price of 40, and marginal cost there is 22. Price is above marginal cost, so some worthwhile trades do not occur.

what a buyer would pay for a 7th unit	$58 - 3(7) = 37$
what a 7th unit costs to make	$TC(7) - TC(6) = 263 - 240 = 23$
gain if it were made and sold	$37 - 23 = 14$

And it is not made. To sell it the firm must cut the price to 37 on all seven, giving up \$3 on each of the six — 18 — so the unit brings in only $37 - 18 = 19$ against a cost of 23.

Why it cannot simply be fixed

Force the firm to charge marginal cost at its own quantity: $P = MC = 22$ at $Q = 6$.

revenue	$22 \times 6 = 132$
total cost	$144 + 10(6) + 6^2 = 144 + 60 + 36 = 240$
result	$132 - 240 = -108$

The firm closes. Marginal cost pricing does not cover the fixed 144 — and it never can while average cost is still falling, because MC **lies below** ATC **everywhere to the left of** $Q = 12$**.**

The product-variety externality

Product-variety externality. When a new firm enters with a product of its own, consumers who prefer it gain surplus that the entrant does not collect.

Why the entrant does not collect it. It charges one price. Buyers who would have paid far more for *this particular* blend keep the difference, and that gain never appears in the entrant's calculation of whether to open.

So entry brings a benefit the entrant ignores. On this count alone, there is **too little** entry.

The business-stealing externality

Business-stealing externality. When a new firm enters, it takes customers from firms already in the market, and their loss is no part of its calculation.

This is the shift in Part 5, seen from outside. Each entrant pushes every incumbent's demand curve down. The entrant counts its own profit; it does not count the profit it removes from others.

So entry imposes a cost the entrant ignores. On this count alone, there is **too much** entry.

Two effects, opposite directions, no verdict

	who bears it	entry is therefore
Product variety — gain to consumers	consumers	too little
Business stealing — loss to rivals	other firms	too much

Economics does not know which is larger. It depends on how much consumers value the extra variety and how much of the entrant's business is genuinely new rather than taken.

So there is no case here for restricting entry, and none for subsidizing it.
The honest answer is that the sign is unknown.

What the model assumes

Four assumptions did real work, and each binds somewhere.

1. **Costs are identical across firms.** Real entrants differ, and zero profit then applies to the *marginal* entrant, not to everyone. Better firms keep a profit indefinitely.
2. **The products are differentiated, but where each sits is given.** Firms in fact *choose* what to make and how to be different, and that choice is most of what they do.
3. **Entry is free and immediate.** Adjustment takes years. Everything called long run here is a destination, not a date.
4. **One period, no strategy.** Nobody reacts to a named rival. With few enough sellers they do, and that needs different tools.

Active Learning 9

Our roaster: $P = 58 - 3Q$, $TC = 144 + 10Q + Q^2$, long-run $Q = 6$, $P = 40$, $MC = 22$.

1. Force $P = MC = 22$ at $Q = 6$. Find revenue, total cost, and the result.
2. Compare that result with the fixed cost. What does the firm do?
3. Which of the two externalities of entry does that calculation say **nothing** about, and why?
4. A town council proposes capping the number of roasters. Using both externalities, say what the model supports.

Four minutes. Parts three and four need a sentence each, not a paragraph.

A little power over price, and none over profit

Because the product is its own, the firm faces a demand curve that slopes down, and sets a price above marginal cost. Because anyone may enter, that price is competed down until it equals average cost and nothing is left over. The two forces settle at a tangency: the firm charges 40, makes 6, earns nothing, and stops short of the 12 at which its costs would be lowest.

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